DATA 4.2

Quotient-Density Compass and Conservative Fallback

Formal architecture, theorem package, experimental evidence, and Hodge research program

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August 17, 2026

Scientific-status note. DATA 4.2 is an experimental ATP architecture, not a claimed solution of the Hodge conjecture. Mathematical statements labeled theorem below are proved from their stated hypotheses. Empirical statements are restricted to the cited frozen experiments. Conjectures are explicitly labeled. The verifier, not the learned controller, is sovereign.

Contents

1 Executive summary

DATA 4.2 names the architecture selected after the DATA 4.1-C controlled-creativity experiment: *formal quotient representation → quotient-density-trained compass → compass-ordered search → conservative exhaustive fallback → formal verification.* Reflection and novelty remain bounded supervisory mechanisms rather than primary proof engines.

The empirical reason for the change is unusually clean. On 500 untouched DATA 4.1-C confirmation targets, the uninformed DATA 3.0 ordering had mean rank 16.010, while the formal compass baseline C_0 had mean rank 8.222. The creativity variants were essentially tied with C_0: fixed 8.214, reflective 8.218, meta 8.214, random 8.220. Thus the strongest observed signal is representation plus compass navigation, not novelty injection. DATA 4.2 promotes that signal and protects it with complete fallback.

2 Formal proof-search setting

Let F be a recursively checkable formal system, Γ a finite or effectively presented hypothesis set, and φ a target WFF. A proof-search graph is $G = (V, E)$ with initial state v_0 and certificate set C. A path $v_0 \to \cdots \to c \in C$ is accepted only when its extracted proof object is accepted by the native verifier.

Definition 2.1 (Theorem ocean). *For budget B, the bounded theorem ocean is $\mathcal{O}_B(v_0) = \{v : v \text{ is reachable from } v_0 \text{ within } B\}$. The unbounded ocean is the effective union of these layers.*

Definition 2.2 (Solved DAG). *A solved DAG is retained verified search history containing a certificate path and enough predecessor/successor information to label useful local navigation decisions.*

Definition 2.3 (Proof compass). *A compass is a score/ranking rule $K(v, u)$ for legal successors $u \in \Gamma(v)$. A ranking compass reorders candidates but does not delete them.*

3 Quotient before compass

Raw proof syntax contains enormous redundancy. DATA 4.2 therefore chooses an effectively recognizable verifier-respecting equivalence or canonicalization $q : V \to \mathcal{Q}$ before learning. The quotient must be operational: unlike the semantic statement "all proofs of φ are equivalent as proofs of φ," it must be recognizable without already solving the target.

Definition 3.1 (Proof-search congruence). *An equivalence relation $\sim$ on V is a proof-search congruence for C if $v \sim w$ preserves certificate reachability and legal transitions descend consistently to equivalence classes.*

Proposition 3.2 (Quotient-first reachability). *If $\sim$ is a proof-search congruence, complete search on $V/\!\sim$ preserves reachability of C.*

Proof. A certificate-reaching path projects to quotient classes. Conversely, by transition compatibility, every projected legal transition has a representative transition preserving the reachability predicate. □

Theorem 3.3 (Quotient Compression Advantage). *Suppose a finite search layer contains N raw states but only K operational quotient classes, and one representative per class suffices for the chosen reachability objective. Then exhaustive representative search requires at most K state examinations instead of N; the raw-to-quotient examination factor is N/K.*

Proof. Immediate from searching one representative for each of the K classes rather than all N members. □

4 Quotient-density training

For target quotient state q, let $B(q)$ be a local quotient neighborhood and let $p(q)$ be the training-distribution mass of solved examples in $B(q)$. DATA 4.2 treats this *formal solved-DAG density* as a controlled training variable. It is distinct from document density, token frequency, or physics-corpus density.

Theorem 4.1 (Quotient-Density Calibration Theorem)**.** *Assume m IID solved training states; local quotient constancy in $B(q)$; one-hit transfer; and uniform random ranking when $B(q)$ is unseen. With K candidates,*

$$\mathbb{E}[R \mid q] = 1 + \frac{K-1}{2}(1-p(q))^m.$$

For $m > 0$, $K > 1$, and $p < 1$, this expected rank decreases strictly with p.

Proof. The probability of no local training hit is $(1-p)^m$. Rank is 1 after a hit and has expectation $(K+1)/2$ otherwise. Differentiate. □

Corollary 4.2 (Diminishing density returns)**.** *For $m > 1$, the marginal rank benefit has magnitude $\frac{m(K-1)}{2}(1-p)^{m-1}$, which decreases with p.*

Theorem 4.3 (Density-Conditional Navigation Advantage)**.** *Under the same hypotheses, the gain over uniform random ordering is*

$$G(p) = \frac{K-1}{2}\left[1-(1-p)^m\right],$$

which is strictly increasing in p for $0 \le p < 1$.

Proof. Subtract the calibration formula from $(K+1)/2$ and differentiate. □

5 Compass learnability and navigation

Theorem 5.1 (Compass Learnability Theorem, finite-class IID form)**.** *Let $\mathcal{H}$ be a finite class of rankers on K candidates. If IID training/evaluation share a distribution and some $h^* \in \mathcal{H}$ has Top-1 accuracy at least $1/K + 2\gamma$, then empirical Top-1 maximization with*

$$m \ge \frac{2}{\gamma^2} \log \frac{2|\mathcal{H}|}{\delta}$$

returns, with probability at least $1-\delta$, a compass with accuracy at least $1/K + \gamma$.

Proof. Apply Hoeffding's inequality to each ranker and a union bound over $\mathcal{H}$; empirical optimality transfers the 2γ population margin to a γ guaranteed margin. □

Theorem 5.2 (Navigation Advantage Theorem)**.** *If instead some h^* has normalized expected rank loss at most $1/2 - 2\gamma$, empirical rank-loss minimization under the same sample bound returns, with probability $1-\delta$, a compass satisfying*

$$\mathbb{E}[R_{\hat{h}}] \le \frac{K+1}{2} - \gamma(K-1) < \frac{K+1}{2}.$$

Proposition 5.3 (No-Signal Impossibility)**.** *If the visible quotient representation is statistically independent of the identity of every proof-successor, no learner using only that representation can have systematic out-of-sample navigation advantage over the corresponding prior-optimal uninformed policy.*

Proof. Conditioning on the visible representation leaves the successor distribution unchanged, so no measurable function of that representation can extract missing predictive information. □

6 Conservative fallback and the horizon

The compass is allowed to be wrong. DATA 4.2 therefore distinguishes *priority* from *deletion.* High-confidence compass regions are searched first; unvisited legal candidates remain on an exhaustive schedule.

Theorem 6.1 (Fallback Safety)**.** *If compass guidance only permutes a finite candidate set and exhaustive fallback eventually traverses the entire permutation, eventual finite coverage is identical to baseline exhaustive search.*

Proof. A permutation preserves membership; exhaustive traversal visits every candidate. □

Definition 6.2 (Weighted Proof Horizon)**.** *Let $w : E \to \mathbb{Q}_{\geq 0}$ be exact edge costs. For the class $[\varphi]$ of verifier-accepted proofs of φ, define*

$$H_w(\Gamma, \varphi) = \inf_{P \in [\varphi]} w(P).$$

When the infimum is attained, H_w is the minimum proof cost. With unit weights, it is shortest proof length.

Theorem 6.3 (Horizon Certification Theorem)**.** *Assume nonnegative exact costs and an exhaustive search ordered so that, before terminating at incumbent proof P^* of cost c, every search state capable of yielding a proof of cost $< c$ has been exhausted or soundly ruled out. Then $c = H_w(\Gamma, \varphi)$.*

Proof. The verifier establishes a proof of cost c, so $H_w \leq c$. A cheaper proof would contradict exhaustive exhaustion or sound ruling-out below c. Thus $H_w \geq c$. □

Corollary 6.4 (Compass-Horizon Compatibility)**.** *A fallible compass can accelerate discovery without corrupting a horizon certificate provided fallback eventually exhausts all lower-cost regions required by the Horizon Certification Theorem.*

7 Hilbert-ordered theorem search

DATA's Hilbert component is a deterministic fair ordering device, not a proof oracle. Encode a bounded feature/search box into $[0, 1]^n$ and use finite-order Hilbert approximants H_r to enumerate cells while preserving locality better than an arbitrary permutation. Candidate proof states assigned to cells are visited according to increasing Hilbert parameter, with deterministic ties.

For a grid with $M = 2^{rn}$ cells, a target cell uniformly distributed in the ordering has blind mean rank $(M + 1)/2$. A compass can prioritize Hilbert blocks; fallback retains the untouched Hilbert enumeration. In an unbounded effective search, dovetail over resolution/depth/resource shells so every finite encoded proof receives a finite visit time.

Theorem 7.1 (Hilbert-Fallback Completeness, conditional form)**.** *Suppose every finite verifier-checkable proof has an effective encoding in some finite shell S_j, each S_j has a finite Hilbert ordering, and the outer scheduler fairly revisits every shell. Then every existing finite proof is eventually enumerated, regardless of any finite amount of compass-first work inserted between fallback steps.*

Proof. A finite proof belongs to some shell S_j and has finite rank there. Fair scheduling gives S_j infinitely many fallback turns; after finitely many such turns its rank is reached. □

8 Headroom and bounded reflection

Definition 8.1 (Compass headroom)**.** *For rank $R \geq 1$, define headroom above perfect rank as $R - 1$ (or its normalized expectation). Reflection can only improve navigation to the extent that nonzero headroom remains.*

Theorem 8.2 (Headroom Collapse)**.** *If a compass ranks the correct candidate first on every episode, no controller that merely reorders the same candidate set can strictly improve rank.*

Proof. Rank 1 is minimal. □

Theorem 8.3 (Controller Redundancy under Sufficient Statistics)**.** *If the quotient state is a sufficient statistic for the optimal successor and the base compass implements the Bayes-optimal ranking from that statistic, any additional controller observing no extra information cannot improve Bayes risk.*

Proof. By sufficiency, conditioning on additional variables available to the controller does not change the conditional law relevant to the optimal decision. □

These results explain why DATA 4.1-C could show strong compass performance yet essentially zero average gain from meta-creativity: a good representation can consume the available navigation headroom.

9 DATA 4.2 architecture

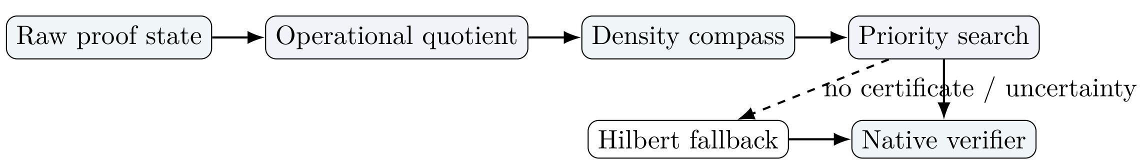

Figure 1: DATA 4.2: representation before reflection, guidance before fallback, verifier sovereignty throughout.

Operationally: freeze formal system, target masking, costs, budgets, and verifier; canonicalize to quotient states; train only on retained solved DAGs with explicit density shells; score successors; search high-ranked candidates first; estimate uncertainty/stagnation; interleave exhaustive Hilbert/best-first fallback; accept only verifier certificates; retain new solved DAGs only under a preregistered update protocol.

10 Current evidence and what it denies

DATA 4.1-C supports a strong representation/navigation effect on its frozen benchmark but not a meaningful creativity effect. Separately, DATA 4.1-Q did *not* justify "density never helps." Formal quotient density was associated with better navigation, while the tested physics-density concentration mechanism failed its strong preregistered advantage and shell-concentration claims. DATA 4.2 therefore studies solved-DAG/formal-quotient density rather than physics-document density.

11 Conjectures for DATA 4.2

Conjecture 11.1 (Quotient-Compass Dominance). *For theorem families possessing a stable operational quotient, quotient-first compass search has lower expected verifier calls than comparably trained raw-state compass search at fixed fallback guarantees.*

Conjecture 11.2 (Quotient Learnability). *There exist natural formal theorem families for which quotienting strictly lowers the sample complexity needed to obtain a fixed navigation advantage.*

Conjecture 11.3 (Metacognitive Headroom Principle). *The maximum expected gain from a reflective controller is bounded by a function f of residual base-compass headroom with $f(0) = 0$.*

Conjecture 11.4 (Reflective Crossover). *Reflection becomes beneficial primarily in regions where compass uncertainty, distribution shift, or successor instability exceeds a measurable threshold.*

Conjecture 11.5 (Representation-Control Substitution). *As an operational quotient approaches a sufficient statistic for proof navigation, the marginal value of increasingly elaborate control decreases.*

Conjecture 11.6 (Density Scaling Law). *For fixed theorem family, quotient, learner, and test distribution, navigation gain is a reproducible saturating function of retained solved-DAG density. A first candidate is $G(p) = \frac{K-1}{2}[1 - (1 - p)^m]$, with realistic data expected to require a softened analogue.*

Conjecture 11.7 (Successor Robustness). *A compass trained on solved DAGs generalizes reliably only when useful successor classes are stable under the quotient and under small admissible perturbations of visible state.*

12 Imaginative theorem forms retained from compass theory

Earlier imaginative ideas become rigorous conditionally. Safe Compass Shrinkage says that if every solvable state retains at least one proof-preserving successor and retained branching is $k < b$, a depth-d tree bound falls from $1 + b + \cdots + b^d$ to $1 + k + \cdots + k^d$. Uniform safe retention $k \leq \rho b$ gives a depth-d leaf ratio at most ρ^d. Combined with average quotient compression factor $m > 1$, the leading-order reduction is approximately $m^{-1}\rho^d$ under the tree/overlap approximation. The difficult empirical question is whether a learned compass satisfies the safe-retention hypotheses.

13 Research program toward the Hodge conjecture

DATA 4.2 changes the Hodge program from "add more domain text" to "build and test formal neighborhoods around certified Hodge-adjacent lemmas." No result here settles the Hodge conjecture. The target is a ladder of verifier-checkable bridge statements.

1. **Hodge-adjacent solved-DAG corpus.** Formalize and retain proofs for linear algebra, inner-product spaces, exterior powers, orientations, Hodge star identities, adjoints, harmonic representatives, finite-dimensional decompositions, and algebraic-cycle/cohomology bridge lemmas genuinely needed by the formalization.

2. **Hodge operational quotients.** Quotient coordinate artifacts, basis renamings, commuting independent steps, canonical linear-map normal forms, duality symmetries, and verified DAG identities when these preserve target reachability.

3. **Density-shell experiment.** Freeze a Hodge-adjacent test set and train on nested solved-DAG shells such as 10, 20, 40, 80. Measure verifier calls, hidden-support rank, Top-k recall, fallback fraction, and proof survival. Fit saturation only after preregistration.

4. **Bridge-focused compass.** Continue the finite-basis/covariant Hodge-star bridge program: rank formal lemmas connecting local coordinate constructions to intrinsic bundle-level statements. Failures become labeled boundary data rather than informal near-proofs.

5. **Quotient stability tests.** Equivalent representatives should induce compatible successor rankings and must not erase obstructions, especially when moving from local/trivialized to global geometric statements.

6. **Horizon measurement.** For solved Hodge-adjacent lemmas, estimate or certify proof horizons under fixed costs. Test whether quotient density reduces discovery cost while leaving certified horizon unchanged.

7. **Negative-result harvesting.** When a compass confidently fails, search for the missing invariant or quotient coordinate. DATA 4.1-C suggests representation repair may be more valuable than generic creativity.

A new *Hodge Bridge Density Conjecture*: within a frozen family of formal Hodge-adjacent bridge problems, increasing solved-DAG density in the correct operational quotient neighborhood yields a monotone, saturating increase in compass navigation advantage, while unrelated physics-document density has no guaranteed effect. This is testable and much weaker than the Hodge conjecture itself.

14 What would justify DATA 5.0?

DATA 4.2 remains 4.x because it consolidates existing mechanisms. A plausible DATA 5.0 threshold is learned construction or revision of verifier-respecting quotients; a reproducible cross-family density/navigation law used by an autonomous controller; or another architectural discontinuity demonstrated on held-out formal problems. Merely adding a larger model or more novelty is not enough.

15 Reproducibility protocol

Freeze before testing: repository commit; formal library and verifier version; target set; train/test split; quotient map; density-shell construction; candidate generator; compass class and hyperparameters; fallback schedule; random seeds; cost model; stopping rules; and statistical hypotheses. Store certificates, predecessor DAGs, ranks, verifier-call counts, fallback events, failures, and environment metadata. Report negative results with the same prominence as positive results.

16 Conclusion

DATA 4.2 is the current synthesis: **quotient first, learn the compass from solved formal structure, exploit local solved-DAG density, search aggressively but conservatively, fall back fairly, and let the verifier decide**. Hilbert ordering supplies a deterministic locality-preserving fallback enumeration; the Proof Horizon separates discovery effort from minimum proof cost; quotient compass theory explains when representation can reduce navigation burden; and the Hodge research program becomes frozen, verifier-checkable bridge experiments rather than an assertion that heuristic success constitutes a proof.

AI integrity statement

This report was prepared with substantial assistance from OpenAI's ChatGPT in mathematical organization, theorem restatement, proof checking at the level shown, technical writing, and document production. Brian Tenneson served as research consultant and directed the architecture, experimental questions, naming, and publication requirements. All theorem claims are intended to be read only with their stated hypotheses; all empirical claims should be checked against archived experimental artifacts and native verifier outputs.

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